

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

September 10, 2025

Outline

- 1 SoPS Lecture 5
 - Battleship
 - Probability and Expectation Values
 - Wrapping up

Outline

1 SoPS Lecture 5

- Battleship
- Probability and Expectation Values
- Wrapping up

Chomp Review

Last time

Chomp Review

Last time

- we explored 3×3 Chomp

Chomp Review

Last time

- we explored 3×3 Chomp
- we learned **proof by cases**

Chomp Review

Last time

- we explored 3×3 Chomp
- we learned **proof by cases**
- we learned the **Strong Law of Small Numbers** and saw examples

Battleship Introduction



Activity (30 minutes)

- review the rules of Battleship
- split into pairs and play some games!
- can you observe any useful strategies?

Battleship Discussion

Question 1

How does the game Battleship differ from other games we have played in this class. How is it similar?

Question 2

What mathematical tools could be useful in exploring the game Battleship?

Outline

1

SoPS Lecture 5

- Battleship
- Probability and Expectation Values
- Wrapping up

Probability Intro

Vocabulary

In **probability theory** we study **experiments**, like flipping a coin, rolling a die, or drawing a card.

- the set Ω of all possible outcomes is the **sample space**
- subsets $E \subseteq \Omega$ are called **events**
- the **probability** of an event E is

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}$$

Probability Example

Rolling a Die

Experiment: roll a six-sided die, the sample space is:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- the event of “getting a 3” is $\{3\}$

$$\mathbb{P}(\{3\}) = \frac{|\{3\}|}{|\Omega|} = \frac{1}{6}.$$

- the event of “getting an even number” is $\{2, 4, 6\}$

$$\mathbb{P}(\{2, 4, 6\}) = \frac{|\{2, 4, 6\}|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}.$$

Probability Example

Drawing a Card

Experiment: draw a card from a standard deck

- what is the sample space?

$$\Omega = \{(2, \heartsuit), (3, \heartsuit), (4, \heartsuit), \dots, (K, \clubsuit), (A, \clubsuit)\}$$

- what is the event "getting a face card in hearts"?

$$E = \{(J, \heartsuit), (Q, \heartsuit), (K, \heartsuit)\}$$

- what is the probability of this event?

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{3}{52}.$$

Probability Meaning

Real-world interpretation

- perform the same experiment a thousand times
- record how many times the outcome is in E
- the fraction of times the outcome is in E should be close to $\mathbb{P}(E)$
- more experiments $\rightarrow$ closer to $\mathbb{P}(E)$

Probability Experiment

Activity (5 minutes)

Roll a die and keep track of

- how many rolls you do
- how many times the outcome is even

Gather the data

Add up the results in your group and be prepared to share!

Expectation Value

Vocabulary

A **value function** $f(x)$ assigns a value to each possible outcome of an experiment

Examples

- how much money we win/lose in a bet
- the position of a random walker

Expectation Value

Start with a sample space

$$\Omega = \{a_1, a_2, a_3, \dots, a_n\}.$$

Vocabulary

A **expectation value** of a **value function** $f(x)$ is

$$\mathbb{E}(f) = f(a_1)\mathbb{P}(\{a_1\}) + f(a_2)\mathbb{P}(\{a_2\}) + \dots + f(a_n)\mathbb{P}(\{a_n\})$$

Meaning

Represents the average value we obtain if we do thousands of experiments

Expectation Value Example

Gambling with dice

Suppose we play a game where we roll a die. If the outcome is a 6, then we win 4 dollars. Otherwise, we lose a dollar.

- value function

$$f(1) = -1, f(2) = -1, f(3) = -1, f(4) = -1, f(5) = -1, f(6) = 4$$

- expectation value

$$\mathbb{E}(f) = (-1) \cdot \frac{1}{6} + (-1) \cdot \frac{1}{6} + (-1) \cdot \frac{1}{6} + (-1) \cdot \frac{1}{6} + (-1) \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} = -\frac{1}{6}$$

- if we play a thousand games, we'll lose an average of 17 cents per game or about 166.67 dollars total.

Expectations in Battleship

Suppose we play mini-Battleship on a 5×5 board, with a single 2×1 destroyer which is placed randomly.

[

Question 1] How many things are in the sample space?

[

Question 2] What is the probability of a ship being in each spot?

Expectations in Battleship

Suppose we play mini-Battleship on a 5×5 board, with a single 3×1 submarine which is placed randomly.

[

Question 3] How many things are in the sample space?

[

Question 4] What is the probability of a ship being in each spot?

Expectations in Battleship

Suppose we take our first shot at the center and we miss.

[

Question 5] How do the probabilities change?

Outline

- 1 SoPS Lecture 5
 - Battleship
 - Probability and Expectation Values
 - Wrapping up

Discussion

Discussion

The presence of the unknown

In Battleship, we no longer have **perfect information**. There are things one player knows that another does not. This makes tools like probability

Discussion

The presence of the unknown

In Battleship, we no longer have **perfect information**. There are things one player knows that another does not. This makes tools like probability

Question 1

What does probability mean in your own words?

Discussion

The presence of the unknown

In Battleship, we no longer have **perfect information**. There are things one player knows that another does not. This makes tools like probability

Question 1

What does probability mean in your own words?

Question 2

How can we use probability to play Battleship better?

Question of the Day

Question of the Day

Puzzle

Suppose we play mini-Battleship on a 5×5 board. How many ways are there to place a 2×1 Destroyer and a 3×1 submarine on the board **simultaneously**.

Final thoughts

Reflection

Final thoughts

Reflection

- Can you think of any other games where probability may be an important tool? Describe one.

Final thoughts

Reflection

- Can you think of any other games where probability may be an important tool? Describe one.
- What strategies would you suggest for placing the ships in Battleship? What about firing the shots?

Submit your responses by **Tonight!**